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Wireless endoscope, wireless endoscope apparatus and illumination control method

Abstract

A wireless endoscope includes: a wireless receiving circuit configured to receive a wirelessly transmitted voice; a first processor configured to process the voice received by the wireless receiving circuit; a light source apparatus configured to generate illumination light to illuminate a subject; an image pickup sensor configured to pick up an image of the subject; a wireless transmission circuit configured to wirelessly transmit an image pickup result of the image pickup sensor; and a second processor configured to control the light source apparatus based on a processing result of the first processor.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of PCT/R2019/014744 filed on Apr. 3, 2019, the entire contents of which are incorporated herein by this reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a wireless endoscope, a wireless endoscope apparatus and an illumination control method that are suitable for vocal cord observation.

2. Description of the Related Art

(2) Conventionally, an endoscope system has been widely used. In the endoscope system, an elongated endoscope is inserted into a body cavity or the like, and observation or various treatments of a site to be examined are performed. An endoscope image obtained by an image pickup device of the endoscope is transmitted to a video processor that performs signal processing. The video processor performs the signal processing of the image from the endoscope, and supplies the image to a monitor for display or supplies the image to a recording apparatus for recording.

(3) A scope cable is used for transmitting the endoscope image from the endoscope to the video processor. However, by the scope cable, the movement range of the endoscope can be restricted or the operability can be impaired. Further, the scope cable can be tangled in another cable, causing a failure such as disconnection. Hence, in recent years, there has been developed a wireless endoscope that is equipped with a rechargeable battery and that wirelessly transmits the endoscope image to the processor and the like.

(4) In some cases, vocal cord observation is performed by the endoscope. For the observation of vocal cord vibration, stroboscopic observation is employed. In the stroboscopic observation, an observation image is obtained using a so-called stroboscopic effect by which the vocal cords look as if the vocal cords were slowly vibrating by blinking stroboscopic light with a period slightly different from the period of the vocal cord vibration.

(5) In the stroboscopic observation with the endoscope, for example, voice analysis is performed based on an acquired sound, and light emission (stroboscopic light emission) is performed at light emission timings decided from the result of the voice analysis (see Japanese Patent Application Laid-Open Publication No. 2000-300514, for example). Accordingly, a microphone for acquiring the voice of a patient is necessary for the stroboscopic observation of the vocal cords.

(6) Note that the wireless endoscope can be employed as an endoscope that allows the observation of vocal cords in the stroboscopic observation.

SUMMARY OF THE INVENTION

(7) A wireless endoscope according to an aspect of the present invention includes: a wireless receiving circuit configured to receive a wirelessly transmitted voice; a first processor configured to process the voice received by the wireless receiving circuit; a light source apparatus configured to generate illumination light to illuminate a subject; an image pickup sensor configured to pick up an image of the subject; a wireless transmission circuit configured to wirelessly transmit an image pickup result of the image pickup sensor; and a second processor configured to control the light source apparatus based on a processing result of the first processor.

(8) A wireless endoscope apparatus according to an aspect of the present invention includes a wireless endoscope and a voice detection unit, the voice detection unit including: a microphone configured to detect a voice; a voice modulation circuit configured to modulate the voice detected by the microphone; and a first wireless transmission circuit configured to wirelessly transmit a modulation result of the voice modulation circuit, the wireless endoscope including: a wireless receiving circuit configured to receive the voice wirelessly sent from the first wireless transmission circuit; a first processor configured to process the voice received by the wireless receiving circuit; a light source apparatus configured to generate illumination light to illuminate a subject; an image pickup sensor configured to pick up an image of the subject; a second wireless transmission circuit configured to wirelessly transmit an image pickup result of the image pickup sensor; and a second processor configured to control the light source apparatus based on a processing result of the first processor.

(9) An illumination control method according to an aspect of the present invention is an illumination control method for controlling illumination light of a wireless endoscope, in which: a voice detection unit detects a voice, modulates the detected voice, and wirelessly transmits a signal for the modulated voice; and a wireless endoscope receives the wirelessly transmitted voice, and controls light emission by giving a processing result of the wirelessly received voice, to a light source apparatus configured to generate illumination light to illuminate a subject.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram showing a circuit configuration of a principal part of a wireless endoscope apparatus according to a first embodiment of the present invention;
(2) FIG. **2** is an explanatory diagram showing an outline of a whole configuration of an endoscope system including the wireless endoscope apparatus;
(3) FIG. **3** is an explanatory diagram for explaining an attachment position of a microphone unit on a wireless endoscope;
(4) FIG. **4** is a flowchart for explaining an operation of the first embodiment;
(5) FIG. **5** is an explanatory diagram for explaining a manner of stroboscopic observation in the embodiment;
(6) FIG. **6** is a block diagram showing a second embodiment of the present invention; and
(7) FIG. **7** is a flowchart for explaining an operation of the second embodiment.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(8) Embodiments of the present invention will be explained below in detail, with reference to the drawings.
First Embodiment
(9) FIG. **1** is a block diagram showing a circuit configuration of a principal part of a wireless endoscope apparatus according to a first embodiment of the present invention. FIG. **2** is an explanatory diagram showing an outline of a whole configuration of an endoscope system including the wireless endoscope apparatus. FIG. **3** is an explanatory diagram for explaining an attachment position of a microphone unit on a wireless endoscope. In the embodiment, the microphone unit and the wireless endoscope are wirelessly connected, and stroboscopic light emission is controlled by detection of attachment of the microphone unit, so that stroboscopic observation using the wireless endoscope can be performed.
(10) First, an outline of the endoscope system employing a wireless endoscope will be explained with reference to FIG. **2** and FIG. **3**. The endoscope system in FIG. **2** includes the wireless endoscope **1**, a video processor **5** and a monitor **6**. Various medical instruments and the monitor **6** are disposed on a cart **7**. Further, the video processor **5** connected with a wireless section **8** is placed on the cart **7**. Note that apparatuses such as an electric cautery apparatus, an insufflation apparatus and a video recorder, and a gas cylinder filled with carbon dioxide also are placed on the cart **7** as medical instruments, for example.
(11) The wireless endoscope **1** can perform a photographing operation for ordinary endoscope observation when an unillustrated battery is mounted, and has a wireless configuration of being wirelessly connected with the video processor **5**.
(12) The wireless endoscope **1** is configured by an endoscope body including an insertion section **2** on a distal end side and including an operation section **3** on a proximal end side. An unillustrated image pickup section as an image pickup sensor is arranged at a distal end portion of the insertion section **2**. The image pickup section includes an image pickup device configured by a CCD, a CMOS sensor or the like. Further, an unillustrated illumination section configured to generate illumination light for illuminating a photographic subject is provided in the insertion section **2**, The

illumination section as an in-scope light source irradiates the photographic subject with the generated light as the illumination light, through an unillustrated lens al a distal end of the insertion section **2**.

(13) Return light from the photographic subject enters an unillustrated observation lens at the distal end of the insertion section **2**, and forms an image on an image pickup surface of the image pickup section. The image pickup section obtains a. pickup image based on a photographic subject optical image, by photoelectric conversion. The image pickup section transmits the pickup image to an unillustrated substrate in the operation section **3**, through an unillustrated signal wire in the insertion section **2**. The unillustrated substrate provided in the operation section **3** is equipped with various circuits including an unillustrated image processing circuit and a communication circuit configured to perform signal transmission, and by the circuits, the pickup image is wirelessly transmitted to the wireless section **8**. Note that a plurality of endoscope switches **3***b* are arranged on an operation surface **3***a* of the operation section **3** and operation signals based on operations of the endoscope switches **3***b* are also transmitted to the wireless section **8**.

(14) The communication circuit incorporated in the wireless endoscope **1** can transmit and receive not only the pickup image but also drive signals and a variety of setting information, for the wireless section **8**. By the drive signals, the image pickup device and the illumination section arranged in the insertion section **2** are driven.

(15) Note that although the image pickup section and the illumination section have been explained as an image pickup section and an illumination section that are provided at the distal end of the insertion section **2**, the image pickup section may be provided on the side of the operation section **3** as in a case of a camera head, a light source may be provided in the operation section **3** or the like and the illumination light may be guided to the distal end of the insertion section **2** by a light guide or the like.

(16) Note that a battery **18** is incorporated in the wireless endoscope **1** and electric power from the battery **18** is supplied to a power source section **19** equipped on the substrate in the operation section **3**.

(17) As shown in FIG. **3**, an unillustrated locking portion for attaching a microphone unit **4** is provided on the operation surface **3***a*. of the operation section **3**. The microphone unit **4** as a voice detection unit can be removably attached on the operation surface **3***a*. At the time of stroboscopic observation, the microphone unit **4** is attached to the locking portion. The microphone unit **4** acquires a voice in a periphery, and transmits the acquired voice to the wireless endoscope **1**.

(18) Next, specific configuration examples of the wireless endoscope **1** and the microphone unit **4** will be explained with reference to FIG. **1**.

(19) In the microphone unit **4**, a control section **41**, a microphone sensor **42**, a voice modulation section **43**, a wireless section **44** and a battery **45** are provided. The control section **41** may be configured by a processor using a CPU, an FPGA or the like, and may control each section by operating in accordance with a program stored in an unillustrated memory. Some or all functions may be realized by an electronic circuit of hardware.

(20) As the microphone sensor **42**, various microphones such as a capacitor type and a dynamic type can be employed. The microphone sensor **42** as a voice detection section acquires the voice in the periphery, and outputs the acquired voice to the voice modulation section **43**. As the voice modulation section **43**, for example, modulation circuits employing digital modulation schemes such as QAM (quadrature amplitude modulation) and various other modulation schemes can be used. The voice modulation section **43** as a voice modulation circuit modulates the voice given from the microphone sensor **42**, and outputs the voice to the wireless section **44**. The wireless section **44** as a first wireless transmission circuit is configured by a communication circuit configured to operate under a predetermined wireless scheme. For example, the wireless section **44** may be capable of performing a short-range wireless communication, and may employ a wireless standard such as Wi-Fi (registered trademark) and Bluetooth (registered trademark). The wireless

section **44** wirelessly transmits the voice from the voice modulation section **43**.
(21) The voice wirelessly transmitted from the wireless section **44** can be received by a wireless section **16** of the wireless endoscope **1**. The wireless section **16** as a wireless receiving circuit is a communication circuit that operates under a wireless standard corresponding to the wireless section **44**. The wireless section **16** receives the voice transmitted from the wireless section **44**, and outputs the received voice to an image processing and control section **11** through a bus **20**.
(22) The image processing and control section **11** (also referred to as a control section **11**, hereinafter) of the wireless endoscope **1** may be configured by a processor using a CPU, an FPGA or the like, and may control each section by operating in accordance with a program stored in an unillustrated memory. Some or all functions may be realized by an electronic circuit of hardware. The control section **11** as a second processor is connected with the bus **20**, and controls each section through the bus **20**.
(23) In the wireless endoscope **1**, an illumination section **12** is provided. The illumination section **12** as a light source apparatus is configured by various light source apparatus such as an LED (light-emitting diode) light source. The illumination section **12** generates the illumination light by being controlled by the image processing and control section **11**. The photographic subject is irradiated with the illumination light from the illumination section **12**, through an unillustrated illumination lens provided at a distal end portion of the insertion section **2**.
(24) The illumination section **12** performs switching between ordinary light emission and stroboscopic light emission by being controlled by the control section **11**. Further, a light emission control signal is given to the illumination section **12** by a voice processing section **15** described later, and the light emission period at the time of the stroboscopic light emission is controlled based on the light emission control signal.
(25) In the wireless endoscope **1**, an image pickup section **13** is provided at a distal end portion of the insertion section **2**. The image pickup section **13** includes an unillustrated image pickup device such as a CCD or a CMOS sensor, and reflected light from the photographic subject enters an image pickup surface through an unillustrated observation lens. The image pickup section **13** obtains a pickup image by photoelectric conversion of a photographic subject optical image that enters the image pickup surface. The image processing and control section **11** drives the image pickup section **13**, and receives the pickup image from the image pickup section **13**, to output the pickup image to a wireless section **14** after performing a predetermined image signal process.
(26) The wireless section **14** as a second wireless transmission circuit is configured by a communication circuit that employs a predetermined wireless standard such as Wi-Fi. The wireless section **14** modulates the pickup image from the image processing and control section **11**, and thereafter wirelessly transmits the pickup image. The signal transmitted from the wireless section **14** is received by the wireless section **8**, and is supplied to the video processor **5**. Further, the wireless section **14** receives various control signals and the like transmitted from the video processor **5** through the wireless section **8**, and gives the control signals and the like to the image processing and control section **11**.
(27) The battery **18** is provided in the wireless endoscope **1**. The battery **18** generates a predetermined electric power, and supplies the electric power to the power source section **19**. The power source section **19** can be configured by a power source circuit configured to generate power source voltage to he used in each section of the wireless endoscope **1** by the electric power from the battery **18**. The power source section **19** supplies the generated power source voltage to each section of the wireless endoscope **1**, by being controlled by the image processing and control section **11** (not illustrated).
(28) In the embodiment, the image processing and control section **11** performs the stroboscopic light emission of the illumination section **12**, only in a case where the microphone unit **4** is removably attached on the operation surface **3***a* or in a case where a voice having a level equal to or higher than a predetermined level is supplied from the microphone unit **4**.

(29) For example, when the wireless communication between the wireless section **16** and the wireless section **44** is established, the image processing and control section **11** may determine that the microphone unit **4** has been attached to the operation surface **3***a*, and may control the illumination section **12** such that the stroboscopic light emission is started. Further, in a case where the receiving level of the voice received by the wireless section **16** exceeds a predetermined threshold when the receiving level of the voice and the threshold are compared, the image processing and control section **11** may determine that a sufficient level of voice has been outputted from the microphone unit **4**. and may control the illumination section **12** such that the stroboscopic light emission is started.
(30) The voice received by the wireless section **16** is supplied to the voice processing section **15** as a first processor. The voice processing section **15** as the voice processing section may be configured by a processor using a CPU, an FPGA or the like, and may control each section by operating in accordance with a program stored in an unillustrated memory. Some or all functions may be realized by an electronic circuit of hardware. The voice processing section **15** evaluates the voice frequency of the inputted voice by voice analysis, generates the light emission control signal based on the evaluated voice frequency, and outputs the light emission control signal to the illumination section **12**. The illumination section **12** performs the stroboscopic light emission with a period based on the light emission control signal.
(31) The voice processing section **15** is a circuit section necessary for the stroboscopic observation. Hence, in the embodiment, to reduce electric power consumption, the image processing and control section **11** may set a time period during which the wireless communication with the wireless section **44** in the wireless section **16** is established or a time period during which the voice receiving level in the wireless section exceeds the threshold, as a stroboscopic observation time period, and may control the power source section **19** such that the power source voltage is supplied to the voice processing section **15** only during the time period.
(32) Next, an operation of the embodiment configured in this way will be explained with reference to FIG. **4** and FIG. **5**. FIG. **4** is a flowchart for explaining the operation of the embodiment. FIG. **5** is an explanatory diagram for explaining a manner of the stroboscopic observation in the embodiment.
(33) As shown in FIG. **5**, in the vocal cord observation, first, the insertion section **2** of the wireless endoscope **1** is inserted into a mouth **25***a* of a subject **25**, such that the distal end of the insertion section **2** reaches a position allowing the vocal cord observation. During the insertion, an ordinary light observation with ordinary light is performed such that a state at the time of the insertion can he observed (step S**1** in FIG. **4**). In other words, the image processing and control section **11** controls the illumination section **12**, such that the ordinary light is emitted (step S**2**).
(34) The image pickup section **13** performs the photoelectric conversion of the return light from the subject **25** illuminated by the illumination section **12**, arid outputs the pickup image to the image processing and control section **11**. The image processing and control section **11** performs the predetermined image signal process to the pickup image, and thereafter transmits the pickup image to the video processor **5** through the wireless section **14**. The video processor **5** generates an image based on the pickup image, and outputs the image to the monitor **6**. Thus, the pickup image at the time of the insertion is displayed on a display screen of the monitor **6**.
(35) When an operator inserts the insertion section **2** to the position allowing the vocal cord observation, the operator mounts the microphone unit **4** on the operation surface **3***a*. Note that a position where the microphone unit **4** is mounted is shown by a broken line in FIG. **5**. In step S**3**, the image processing and control section **11** determines whether the microphone unit **4** has been mounted. When the microphone unit **4** has been mounted, the wireless communication between the wireless section **44** and the wireless section **16** is established. Information indicating the establishment of the wireless communication is given to the image processing and control section **11**. Thereby, the image processing and control section **11** determines that the microphone unit **4** has

mounted on the operation surface **3***a*, and transfers the process from step S**3** to step S**4**. Steps S**4** to S**8** show processes in the stroboscopic observation time period. The image processing and control section H starts the supply of the power source voltage from the power source section **19** to the voice processing section **15**, and activates the voice processing section **15** (step S**4**). The control section **11** causes the illumination section **12** to start the stroboscopic light emission (step S**5**).

(36) The microphone unit **4** acquires the voice of the subject **25** through the microphone sensor **42** (step S**6**). The voice from the microphone sensor **42** is modulated by the voice modulation section **43**, and is supplied to the wireless section **44**. The wireless section **44** wirelessly transmits the voice. The wireless section **16** of the wireless endoscope **1** receives the voice transmitted from the wireless section **44**, and supplies the voice to the voice processing section **15** through the bus **20**. The voice processing section **15** evaluates the voice frequency from the inputted voice. The voice processing section **15** generates the light emission control signal for causing the illumination section **12** to perform the stroboscopic light emission at a frequency close to the voice frequency, based on the evaluated voice frequency, and supplies the light emission control signal to the illumination section **12** (step S**7**). The illumination section **12** performs the stroboscopic light emission at the light emission frequency based on the light emission control signal (step S**8**).

(37) Thus, the vocal cords are illuminated by the stroboscopic light emission. The image pickup section **13** performs the photoelectric conversion of the return light from the vocal cords illuminated by the stroboscopic light emission, and outputs the pickup image to the image processing and control section **11**. The image processing and control section **11** performs the predetermined image signal process to the pickup image, and thereafter transmits the pickup image to the video processor **5** through the wireless section **14**. The video processor **5** generates an image based on the pickup image, and outputs the pickup image to the monitor **6**. Thus, an image in which the vocal cords slowly vibrate is displayed on a display screen of the monitor **6**, so that the stroboscopic observation can be performed. The processes in steps S**6** to S**8** are repeated until it is detected that the microphone unit **4** has been detached in step S**9**.

(38) The operator detaches the microphone unit **4** for ending. the stroboscopic observation. In step S**9**, the image processing and control section **11** determines whether the microphone unit **4** has been mounted. The wireless communication between the wireless section **44** of the microphone unit **4** and the wireless section **16** is cancelled by the increase in the distance between the two. From the output of the wireless section **16**, the image processing and control section **11** detects that the wireless communication has been cancelled, namely, that the microphone unit **4** has been detached. Then, the image processing and control section **11** controls the illumination section **12** to perform the ordinary light emission control, and ends the stroboscopic observation time period (step S**10**). In step S**11**, the image processing and control section **11** stops the power supply to the voice processing section **15**, and stops the operation of the voice processing section **15**. Thus, the return to the ordinary light observation is performed.

(39) Note that the switching between the stroboscopic observation time period and the ordinary light observation time period may be performed in a state where the microphone unit **4** is mounted, although the switching between the stroboscopic observation time period and the ordinary light observation time period is performed by the determination of the establishment or cancellation of the wireless communication from the attachment or detachment of the microphone unit **4**. For example, the control section **11** may start the stroboscopic light emission when the level of the voice supplied from the microphone unit **4** to the wireless endoscope exceeds a predetermined threshold, and may recover the ordinary light observation when the level of the voice becomes lower than the predetermined threshold.

Second Embodiment

(40) FIG. **6** is a block diagram showing a second embodiment of the present invention. In FIG. **6**, the same component elements as component elements in FIG. **1** are denoted by the same reference characters, and the explanation is omitted. The embodiment allows the management of electric

power in a case of employing a microphone unit that includes no battery.

(41) A microphone unit **40** is different from the microphone unit **4** in FIG. **1**, in that the microphone unit **40** includes a power receiving section **46** as a power receiving circuit instead of the battery **45**. Further, a wireless endoscope **10** is different from the wireless endoscope **1** in FIG. **1**, in that a power transmission section **21** as a power transmission circuit is added. The power transmission section **21** transmits the electric power from the power source section **19** to the microphone unit **40**, by being controlled by the image processing and control section **11**. The power receiving section **46** receives the electric power transmitted from the power transmission section **21**. For example, the power transmission section **21** and the power receiving section **46** are configured by coils, and are electromagnetically coupled, so that the transmission of the electric power can be performed. The power receiving section **46** can receive the supply of the electric power from the power transmission section **21**, by approaching the power transmission section **21**. The power receiving section **46** generates power source voltage to be used in each section in the microphone unit **40**, by the received electric power.

(42) In the embodiment, the power transmission section **21** can detect whether the power transmission section **21** is supplying electric power to the power receiving section **46**. For example, in a case where the power transmission section **21** detects that the power transmission section **21** is supplying electric power to the power receiving section **46**, the power transmission section **21** outputs a detection signal to the image processing and control section **11**. The image processing and control section **11** sets the stroboscopic observation time period based on the detection signal from the power transmission section **21**. Note that the image processing and control section **11** may set the stroboscopic observation time period when the electric consumption amount of the power transmission section **21** exceeds a predetermined reference value. For example, the image processing and control section H may detect that the electric consumption in the microphone unit **40** is increased for transmitting a voice having a level equal to or higher than a predetermined level, and may set the stroboscopic observation time period.

(43) Note that the image processing and control section **11** may set the stroboscopic observation time period by monitoring the electric consumption amount of the power transmission section **21** based on information from the power source section **19**.

(44) Next, an operation of the embodiment configured in this way will be explained with reference to FIG. **7**. FIG. **7** is a flowchart for explaining the operation of the embodiment. In FIG. **7**, the same procedures as procedures in FIG. **4** are denoted by the same reference characters, and the explanation is omitted.

(45) In the embodiment, in step S**3** of FIG. **7**, the mounting of the microphone unit **40** is detected based on determination of whether the power transmission section **21** and the power receiving section **46** have been electromagnetically coupled. When the image processing and control section **11** detects the electromagnetic coupling between the power transmission section **21** and the power receiving section **46**, the image processing and control section **11** determines that the microphone unit **40** has been mounted on the operation surface **3***a*, and transfers the process from step S**3** to step S**4**.

(46) The image processing and control section **11** starts the supply of the power source voltage from the power source section **19** to the voice processing section **15**, and activates the voice processing section **15**. In step S**5**, the image processing and control section **11** starts the stroboscopic light emission. Furthermore, the image processing and control section **11** causes the power transmission section **21** to start the power transmission to the microphone unit **40**, and activates the microphone unit **40** (step S**21**). The microphone unit **40** becomes capable of operating when the power receiving section **46** receives the supply of electric power, and the microphone sensor **42** acquires the voice of the subject by being controlled by the control section **41** (step S**6**). Thus, in the embodiment, the stroboscopic observation can be performed.

(47) In the embodiment, in step S**9** of FIG. **7**, it is detected that the microphone unit **40** has been

detached, based on a determination that the power transmission section **21** and the power receiving section **46** are electromagnetically decoupled. When the image processing and control section **11** detects that the microphone unit **40** has been detached, the image processing and control section **11** performs an ordinary light emission control in step S**10**, and thereafter stops the power transmission to the microphone unit **40** in step S**22**. Note that the power transmission is sometimes stopped automatically when the power receiving section **46** gets away from the power transmission section **21**. In step S**11**, the image processing and control section **11** stops the power supply to the voice processing section **15**, and stops the operation of the voice processing section **15**.
(48) In this way, in the embodiment, it is possible to obtain the same effect as the effect in the first embodiment. Further, in the embodiment, in the case where electric power is supplied from an endoscope processor to the microphone unit, the electric power is supplied to the microphone unit only when the microphone unit is used, so that the increase in electric consumption can be restrained.
(49) In the above embodiment, the example in which the stroboscopic observation time period is set by monitoring the supply of electric power from the power transmission section to the microphone unit, but the stroboscopic observation time period may be set by receiving a voice having a level higher than a predetermined level after the microphone unit starts to operate by receiving the supply of electric power.
(50) Note that the stroboscopic observation time period, a time period during which the illumination section performs the stroboscopic light emission, a time period during which the voice processing section is activated, and a time period during which electric power is supplied to the microphone unit do not always coincide with each other in the above respective embodiments.
(51) In the technologies explained in the specification, for controls explained mainly with the flowcharts, programs can be often set, and are sometimes stored in a recording medium or a recording section. As methods for the recording in the recording medium or the recording section, the recording may be performed at the time of product shipment, a distributed recording medium may be used, or download may be performed through the internet.
(52) Further, for the respective steps in the flowcharts, the execution order may be altered, a plurality of steps may be concurrently executed, or the steps may be in an order that is different for each execution, as long as there is no inconsistency with the characteristics.
(53) Note that parts described as “section” in the embodiments may be configured by combinations of dedicated circuits or a plurality of general-purpose circuits, and may he configured by combinations of processors such as a microcomputer configured to operate in accordance with previously programmed software, and a CPU, or sequencers such as an FPGA, as necessary.
(54) The present invention is not limited only to the above respective embodiments, and can be embodied while component elements are modified without departing from the gist, in the implementation phase. Further, various inventions can be formed by appropriate combinations of a plurality of component elements disclosed in the above respective embodiments. For example, some component elements of all component elements shown in the embodiments may he excluded. Furthermore, component elements across different embodiments may be appropriately combined.

Claims

1. A wireless endoscope comprising: a wireless receiving circuit configured to receive a wirelessly transmitted voice; a first processor configured to process the voice; a light source configured to generate illumination light; an image pickup sensor configured to pick up an image of the subject; and a second processor configured to control the light source based on a processing result of the first processor.
2. A wireless endoscope apparatus comprising a wireless endoscope and a voice detection unit, the voice detection unit including: a microphone configured to acquire a voice; a voice modulation

circuit configured to modulate the voice; and a first wireless transmission circuit configured to wirelessly transmit a modulation result of the voice modulation circuit, the wireless endoscope including: a wireless receiving circuit configured to receive the modulated voice; a first processor configured to process the modulated voice received by the wireless receiving circuit; a light source configured to generate illumination light; an image pickup sensor configured to pick up an image of the subject; a second wireless transmission circuit configured to wirelessly transmit an image pickup result of the image pickup sensor; and a second processor configured to control the light source based on a processing result of the first processor; and wherein the voice detection unit is removably provided on the wireless endoscope.

3. The wireless endoscope apparatus according to claim 2, wherein the second processor is configured to control light emission of the light source based on the processing result.

4. The wireless endoscope apparatus according to claim 2, wherein the voice detection unit includes a battery.

5. The wireless endoscope apparatus according to claim 1, wherein the second processor is configured to put the first processor into an operating state, when the second processor detects that the voice detection unit is mounted on the wireless endoscope, based on a receiving result of the wireless receiving circuit.

6. The wireless endoscope apparatus according to claim 1, wherein the second processor is configured to put the first processor into a non-operating state, when the second processor detects that the voice detection unit is detached from the wireless endoscope, based on a receiving result of the wireless receiving circuit.

7. The wireless endoscope apparatus according to claim 2, wherein the wireless endoscope includes a power transmission circuit, and the voice detection unit includes a power receiving circuit configured to receive electric power from the power transmission circuit.

8. The wireless endoscope apparatus according to claim 7, wherein the voice detection unit is removably provided on the wireless endoscope.

9. The wireless endoscope apparatus according to claim 8, wherein the second processor is configured to put the first processor into an operating state, when the second processor detects that the voice detection unit is mounted on the wireless endoscope, based on a power transmission result of the power transmission circuit.

10. The wireless endoscope apparatus according to claim 8, wherein the second processor is configured to put the first processor into a non-operating state, when the second processor detects that the voice detection unit is detached from the wireless endoscope, based on a power transmission result of the power transmission circuit.

11. The wireless endoscope apparatus according to claim 8, wherein the second processor is changed to an operating state, when the voice detection by the voice detection unit detects that the voice detection unit is mounted on the wireless endoscope, based on a power transmission result of the power transmission circuit.

12. The wireless endoscope apparatus according to claim 8, wherein the second processor is changed to a non-operating state, when the second processor detects that the voice detection unit is detached from the wireless endoscope, based on a power transmission result of the power transmission circuit.

13. An illumination control method comprising: acquiring a voice of a subject through a microphone sensor, picking up an image of vocal cords of the voice acquired through the microphone sensor, evaluating a voice frequency from the voice, generating a light emission control signal based on the voice frequency; and controlling an emission frequency of a light source based on the light emission control signal.

14. The illumination control method according to claim 13, further comprising observing the image of the vocal cords; wherein the light source illuminates the vocal cords under observation.

15. The illumination control method according to claim 13, wherein the acquiring comprises

wirelessly receiving the voice frequency from the microphone sensor.
16. The wireless endoscope according to claim 1, wherein the first processor is configured to evaluate a voice frequency of the voice, wherein the image pickup sensor is configured to pick up an image of vocal cords of the voice.
17. The wireless endoscope according to claim 16, further comprising a wireless transmission circuit configured to wirelessly transmit an image pickup result of the image pick up sensor.
18. The wireless endoscope according to claim 16, wherein the second processor is configured to generate a light emission control signal based on the voice frequency, the second processor is configured to control the light source based on the light emission control signal.
19. The wireless endoscope according to claim 16, wherein the second processor is configured to generate a light emission control signal based on a light emission frequency different from the voice frequency, the second processor is configured to control the light source based on the light emission control signal.